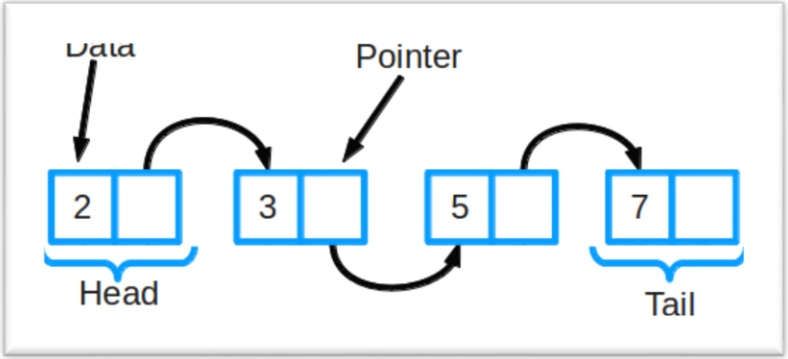
Data Structures And Algorithms 2026



Introduction

ICT2113

Level II – Semester 1

Course Outline

- Lectures – 2 hrs
- Labs – 2 hrs

Assessments

- Continues Assessments

  Three Quizzes : 10%

  Midterm Evaluation : 20%

  Assignment : 10 %

- Final Examination

  Theory : 40%

  Practical :20%

Course Outline

- Lectures
  - Theory
    - Data Structures
    - How to access the data structures
    - Searching and sorting methods
- Tutorials
  - Exercises on data structures and methods
- Labs
  - Hands on experience in how to implement data structures in an application
  - Using C Language in Linux platform
  - Testing and Debugging programs

Course objectives

- ILO1: Describe different types of data structures of common use such as lists, stacks, queues, graphs in programming and their properties.
- ILO2: Construct algorithms to solve problems using data structures.
- ILO3: Describe some useful internal and external sorting algorithms.
- ILO4: Identify graph traversal algorithms, their applications, and the intractable graph problems.

Course Outline

- Arrays
- Stacks
- Queues
- Linked Lists
- Sorting and Searching Methods
- Binary Trees
- Recursion
- Graphs
- Hashing and Heaps





What will we do in this unit?

- Design, implement and test real-world applications
  - Searching
  - Sorting
- Emphasis
  - Programming
  - Algorithm analysis

Prerequisite knowledge…

- C Programming
  - Basic data types
  - Control Structures
  - Functions
  - Arrays
  - Pointers
  - Structures

References

- T. Cormen, C. Leiserson, R. Rivest, and C. Stein, *Introduction to Algorithms, 4th ed.*, MIT Press, 2022. ISBN 978-0262046305.
- N. Karumanchi, *Data Structures and Algorithms Made Easy*, Pearson, 2022. ISBN 978-0471738848.
- *Secure Data Structures and Algorithms with C++: Walls and Mirrors, 8th ed.*, Pearson, 2024. ISBN 978-0138122805.